Task 31: Final Project Video

Multimodal Deepfake Detection for Images, Video and Audio

Team members

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Video link

https://drive.google.com/file/d/1jI6_ZkyNH5SvGP9GO9ITerRVWAO9LoLL/view?usp=sharing

Overall contribution to the project

No.	Task	Assigned to
A. Research and Planning		
1	Problem definition, scope and requirements	Shivansh
2	Literature survey: 86 papers and related-work comparison	Adithya: 43 papers; Shivansh: 22 papers; Vaibhav: 21 papers
3	Dataset access, downloads, licence records and documentation (FF++, Celeb-DF, DFD, ASVspoof, In-the-Wild, LAV-DF, FakeAVCeleb)	Shivansh and Vaibhav
B. Data and Evaluation		
4	Data pipeline: manifests, identity/speaker-separated splits, leakage and duplicate checks	Vaibhav
5	Evaluation protocol: metrics, bootstrap confidence intervals, test guard and experiment ledger	Shivansh
C. Models and Experiments		
6	Preprocessing: frame sampling, face detection and cropping, audio extraction	Shivansh
7	Encoder comparison and first working version (I-0)	Adithya
8	Face-video model: SigLIP fine-tuning, augmentation and ensemble trials (I-1, I-1b, I-1c)	Vaibhav
9	Speech model: silence trimming, augmentation and three-encoder ensemble (I-2 to I-4)	Vaibhav
10	Audio-visual experiments: LAV-DF fusion (I-5, I-6), FakeAVCeleb evaluation and shortcut audits	Adithya
11	Lip-sync module (SyncNet) and validation	Vaibhav
12	Generated-image and generated-video experiments, including modern-generator and Gemini assessment	Shivansh
13	rPPG pulse experiment	Shivansh
14	Xception baseline and final sealed evaluation (T1 to T3)	Vaibhav
D. Software Engineering		
15	Backend: module interface, coordinator, verdict layer and API	Adithya
16	Website and Evaluation Lab frontend	Adithya
17	Provenance modules: C2PA and metadata analysis	Shivansh
18	Testing, quality checks and browser tests	Vaibhav
19	Environment setup, installation instructions and reproducibility	Vaibhav
E. Final Deliverables		
20	Research paper	Shivansh
21	Final technical report	Vaibhav
22	Presentation	Adithya